For the submission of your work:

- Create a folder named **RollNo_Name_DBI202_PaperNo**, e.g. se01245_LongNT_DBI202_01. **Do not** create any subfolder in this folder. All file created will be located in the above folder.

- For each question, you are required to write a database script. Create a file with the name corresponding to the index of the question. For example, **for question 1**, we will create a file named **Q1.sql** and create a file **Q2.sql for question 2**. So, if you do 10 questions, your folder must contain **only** 10 files Q1.sql, Q2.sql, Q3.sql, Q4.sql, Q5.sql, Q6.sql, Q7.sql, Q8.sql, Q9.sql and Q10.sql.

- Do not use any commands having the database name such as create database, alter database, use [database name], GO, EXEC, *etc.*

- Your response must contain only necessary commands for answering the question. Do not include any other command. For example, if you are required to create a trigger/procedure, then your response should contain only commands for creating the corresponding trigger/procedure; all commands for testing the created trigger/procedure are forbidden.

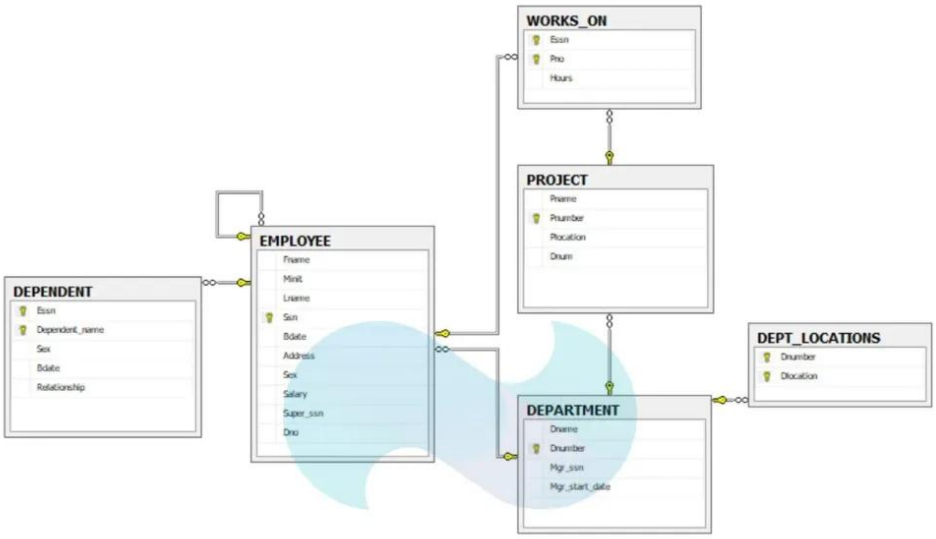
- On completion, import your work by browsing to the above folder.

- Note that:

+ You could use only SQL Server, SQL Server Management Studio, Window explorer, Winrar, Winzip

+ If <u>any</u> of the previous requirements is not respected, your mark will be 0.

From the 2nd question, you should use the database provided in the .sql file which has the following database diagram. Please, run the provided script to create tables and insert data into your database.



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Question 1:

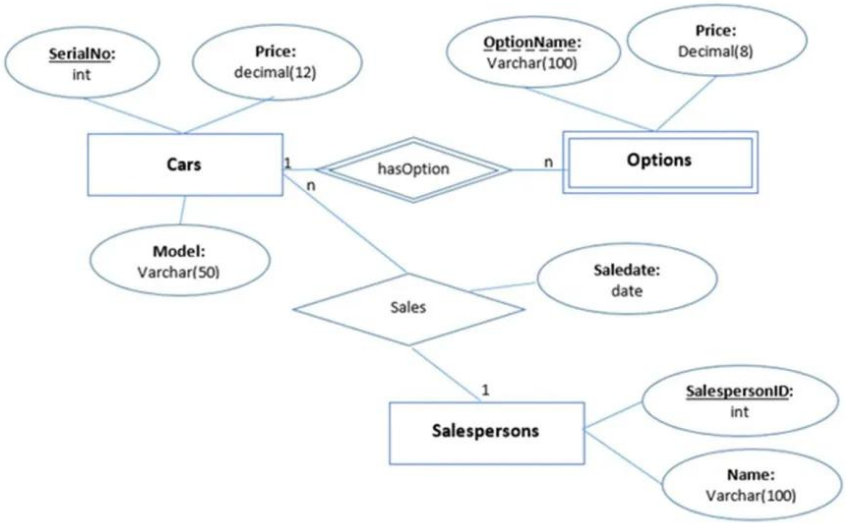
[2 mark] Create one database and then write SQL statements to create, in this database, all tables derived from the ERD given in the following picture with appropriate attributes, primary keys and foreign keys.

NOTE that when creating the SQL commands as request, you MUST keep the name of tables, relationship, attributes and data type of attributes as SAME as given in the given ERD.

Attributes with underline belong to the Primary Key of each entity.

Attributes which reference to the primary key of another table must have the same name as the attributes in the primary key of the referencing table.

When submitting the responses for this question, submit only SQL statements for creating tables with corresponding keys and foreign keys. Do not use "create database", "Alter database", "use database_name" or any statements using database's name in your submission.



Picture 1.1

Question 2:

[1 mark] Write a query to get the average of salary on each department. The result includes two columns: department,average salary.

department	average salary
Administration	40375.000000
Headquaters	63750.000000
Infomation	37500.000000
Research	45428.571428

Picture 2.1

Question 3:

[1 mark] Write a query to get all departments that have more than three employees. Display the result in descending order by number of employees

	Department name	Number of employees
1	Administration	8
2	Research	7
3	Headquaters	4

Picture 3.1

Question 4:

[1 mark] Write a query to retrieve the name of employees (Lname, Fname) whose salary is greater than the salary of all the employees in department 5

	Lname	Fname
1	Tiny	Victoria

Picture 4.1

Question 5:

[1 mark] Write a query to get the department's names that have only performed some projects in 'Houston' and not in 'HCM'

Dname
Administration

Picture 5.1

Question 6:

[1 mark] Write a query to display Fname, Lname, Salary, Dname of the highest paid employee of each department. Display the results as follows:

	Fname	Lname	Salary	Dname
1	Ahn	Seop	80000.00	Research
2	Tim	Rock	60000.00	Administration
3	Huynh	Toan	40000.00	Infomation
4	Victoria	Tiny	85000.00	Headquaters

Picture 6.1

Question 7:

[0.5 mark] Write a query to get all employees that have only performed some projects in 'Stafford' and have not performed any projects in 'Houston'. The result as follow:

Only Stafford not Houston
498546213
567890345
890008978
987987987
999887777

Picture 7.1

Question 8:

[1 mark] Write a procedure named "proc_getRetiredEmployees". This procedure will display all employees who will retire in this year in the department

Note: the retirement age for employees is from 30. When we execute this procedure using the statements as follow: exec proc_getRetiredEmployees 5, the result as follow:

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
Mark	U	Can	111222444	1993-05-12	379 Loop, Metal, TX	M	45000.00	333445555	5
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	35000.00	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000.00	888665555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	30000.00	333445555	5
Ahn	C	Seop	654321789	1988-11-07	456 Green, Houston, TX	M	80000.00	333445555	5
Ramesh	K	Naravan	666884444	1962-09-15	915 Fire Oak, Humble, TX	M	38000.00	333445555	5

Picture 8.1

Question 9:

[1 mark] Create a trigger named "tr_no_delete" for the delete statement on the Dependent table. This trigger does NOT allow you to delete any dependent

The system will display the result "no delete" if you execute the delete statement.

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for example, when you execute the following statement: delete from Dependent.

The system will display the result as in the following figure:

the result
no delete

Picture 9.1

Question 10:

[0.5] Write a query to increase employee salary by an additional 10% if he/she has at least 1 dependent under 18 years old.